Algorithm

Final Exam, June 4, 2025

5:00-8:00 pm

1. (20%) Please answer the following questions:

   (a) Illustrate the operation of Counting-Sort on the array $A = \langle 4, 0, 2, 3, 1, 3, 7, 6, 5, 8, 2 \rangle$ with $k = 8$ and $n = 11$.

   (b) Show that if a node in a binary search tree has two children, then its successor has no left child.

   (c) Show the red-black trees that result after successively inserting the keys $8, 13, 24, 7, 9, 11, 23$ into an initially empty binary search tree.

   (d) Show how Quicksort can be made to run in $O(n \lg n)$ time in the worst case.

2. (15%) A red-black tree is a balanced binary tree that has height $O(\log n)$.

   (a) Show that the subtree rooted at any node x contains at least $2^{bh(x)} - 1$ internal nodes.

   (b) What is the largest possible number of internal nodes in a red-black tree with black height k? What is the smallest possible number?

3. (15%) A subsequence is a sequence that can be derived from another sequence by deleting some elements. Given two sequences $X = \langle x_1, x_2, \ldots, x_m \rangle$ and $Y = \langle y_1, y_2, \ldots, y_n \rangle$, the longest common subsequence problem is to find a maximum-length common subsequence of X and Y.

   (a) Find an LCS of $\langle A, C, D, A, D, A, B \rangle$ and $\langle B, C, D, C, B, A \rangle$.

   (b) Describe an algorithm that solves the longest common subsequence problem in $O(mn)$ time.

4. (20%) Consider inserting the keys $5, 22, 13, 15, 27, 33, 11, 21$ into a hash table of length $m = 13$ using open addressing with the auxiliary hash function $h'(k) = k$. Show the results after the insertion of each number using the following methods.

   (a) Linear hashing.

   (b) Quadratic probing with $c_1 = 1$ and $c_2 = 3$.

   (c) Double hashing with $h_1(k) = k$ and $h_2(k) = 1 + (k \mod (m - 1))$.

   (d) Chaining.

5. (15%) Medians and Order Statistics.

   (a) Design an algorithm to find both the minimum and the maximum of a set of n elements with $3\lfloor n/2 \rfloor$ comparisons in the worse case.

   (b) In the algorithm Select, the input elements are divided into groups of 5. Will the algorithm work in linear time if they are divided into groups of 3?

6. (20%) Consider the knapsack problem in which there are items $i = 1, \ldots, 6$. Item i has weight w_i and value v_i.

i	1	2	3	4	5	6
w_i	1	4	10	13	16	20
v_i	1	6	16	26	40	31

The capacity of the knapsack is 53. You want to maximize the total value you put in the knapsack subject to the constraint that you the total weight carried must be no greater than 53. You can carry at most one unit of each item.

- Solve the problem assuming that you can carry fractional quantities.
- Find the solution to the integer-constrained knapsack problem.